

- 1)
- there exists 1 identity elements in the group, $a \cdot e = a$
 - each elements has its own inverse, $a \cdot a^{-1} = e$
 - associativity, $(a \cdot b) \cdot c = a \cdot (b \cdot c)$
 - closure

abelian groups are commutative, $a \cdot b = b \cdot a$

2) i) identity ✓ 0
inverse ✓ self-inverse

associativity ✗ $(a-b)-c \neq a-(b-c)$

closure ✓ $- : \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}$

ii) identity ✓ 0

inverse ✓ $a^{-1} = -a$

associativity ✓ $(a+b)+c = a+(b+c)$

closure ✓ $+: \mathbb{Q} \times \mathbb{Q} \rightarrow \mathbb{Q}$

$(\mathbb{Z}, +)$, $(\mathbb{Z}_{\text{even}}, +)$

lii) identity ✓ 1

inverse ✗ no inverse for 0

associativity ✓ $abc = a(bc)$

closure ✓ $\times : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$

iv) identity ✓ $-i$

inverse ✓ $a^{-1} = a - 2i$

associativity ✓

closure ✓

$\text{Im}(a), \text{Re}(a) \in \mathbb{Z}$, $\text{Im}(a), \text{Re}(a) \in \mathbb{Q}$

v) identity $\times$

inverse $\times$ $\underline{b} \times \underline{b}^{-1} = e$, but identity doesn't exist

associativity $\times$ let $\underline{b} \parallel e$ $\underline{a} \times \underline{b} \times \underline{c} \neq \underline{a} \times (\underline{b} \times \underline{c})$

closure $\times$ $\underline{b} \parallel e$, $\underline{b} \times \underline{c} = 0$ not in set

vi) identity $\checkmark$ $e = I$

inverse $\checkmark$ $\det(A) \neq 0$

associativity $\checkmark$

closure $\checkmark$ $\det(AB) = \det(A) \det(B) \neq 0$

$(nI, \times)$, $n \in \mathbb{R}$, (rational matrices, $\times$)

vii) identity $\checkmark$ $e = 0$

inverse $\checkmark$ self-inverse

associativity $\times$ $(1 \times 2) \times 3 = 2 \neq 1 \times (2 \times 3)$

closure $\checkmark$

viii) identity $\times$ $2 \times_6 1 = 2$, $2 \times_6 4 = 2$, multiple identities

inverse $\times$ 2^{-1} doesn't exist

associativity $\checkmark$

closure $\times$ $2 \times_6 3 = 0$

ix) identity $\checkmark$ $e=1$

inverse $\checkmark$ $2^{-1}=4$, $3^{-1}=5$, $4^{-1}=2$, $5^{-1}=3$, $6^{-1}=6$

associativity $\checkmark$

closure $\checkmark$

$(\{1, 2, 4\}, \times_7)$, $(\{1, 6\}, \times_7)$

3) $(ab)(cd)$

$$a((bc)d) = a(b(cd)) = (ab)(cd)$$

$$a(b(cd)) = (ab)(cd)$$

$$((ab)c)d = (ab)(cd)$$

$$a((bc)d) = ((ab)c)d = (ab)(cd)$$

a, b, c, d, e

2 groups of 2 : $\left. \begin{array}{l} (ab)(cd)e \\ (ab)(cde) \end{array} \right\} 2$

$$((ab)c)(de)$$

$$(ab)(c(de))$$

by symmetry 6 ways

14 ways in total

1 group of 2 : $\left. \begin{array}{l} ((ab)c)d)e \\ ((a(bc))d)e \\ (a((bc)d))e \\ a((bc)d)e \end{array} \right\} 4$

by symmetry 8 ways

4)

a)

$$AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

$$BA = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 2 \\ 0 & 1 \end{pmatrix}$$

$$(AB)^2 = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 3 \\ 0 & 1 \end{pmatrix}$$

$$A^2 B^2 = \left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \right) \left(\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \right)$$

$$= \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 2 \\ 0 & 1 \end{pmatrix}$$

b) base case $n=1$, true

assume true for $n=k$, $k \in \mathbb{N}$, consider $n=k+1$

$$\begin{aligned}(ab)^{k+1} &= (ab)^k (ab) \\ &= a^k b^k ab \quad , \text{ by commutativity} \\ &= a^{k+1} b^{k+1}\end{aligned}$$

by induction, true for all $n \in \mathbb{N}$

$$abab = aabb$$

$$abab = aab$$

$$a^{-1}aba = a^{-1}aab$$

$$ba = ab$$

$$(ab)^n = a^n b^n, n \in \mathbb{N} \Leftrightarrow a, b \text{ commute}$$

c) $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

the set of 2×2 matrices does not form a group.

$$5) \begin{pmatrix} a & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} b & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} ab & 0 \\ 0 & 0 \end{pmatrix}$$

this is isomorphic to $(\{0\} \setminus \mathbb{R}, \cdot)$

let A, B, C be 2×2 matrices such that $AB = C$

$$\Rightarrow \det(A) \det(B) = \det(C)$$

$$\text{if } \det(A) \neq 0, \det(B) = 0 \Rightarrow \det(C) = 0$$

assume B has an inverse B^{-1} , such that $A = CB^{-1}$

$$\Rightarrow \det(A) = \det(C) \det(B^{-1})$$

$$\det(A) = 0, \text{ contradiction}$$

so to form a group under multiplication, all matrices in the set have non-zero

determinant or zero determinant

$$b) \{nk: k \in \mathbb{Z}\} \subseteq \mathbb{Z}, \quad n \in \mathbb{Z}^+$$

$$e = 0$$

$$(nk)^{-1} = -nk$$

$$nk_1 + nk_2 = n(k_1 + k_2) \in \{nk: k \in \mathbb{Z}\}$$

$$(\{nk: k \in \mathbb{Z}\}, +) \leq (\mathbb{Z}, +)$$

$$k \in m\mathbb{Z}, n\mathbb{Z} \Leftrightarrow k \mid m, n$$

$$\Leftrightarrow (m\mathbb{Z} \cap n\mathbb{Z}, +) = (\text{lcm}(m, n)\mathbb{Z}, +)$$

$$(m\mathbb{Z} \cup n\mathbb{Z}, +) \leq (\mathbb{Z}, +) \Leftrightarrow m \mid n \text{ or } n \mid m$$

7) a) consider $(ABC)_{ij}$, A $m \times n$, B $n \times p$, C $p \times q$

$$\begin{aligned}(AB)C)_{ij} &= \sum_{k=1}^p (AB)_{ik} C_{kj} \\&= \sum_{k=1}^p \left(\sum_{m=1}^n A_{im} B_{mk} \right) C_{kj} \\&= \sum_{k=1}^p \sum_{m=1}^n A_{im} B_{mk} C_{kj}\end{aligned}$$

$$\begin{aligned}(A(BC))_{ij} &= \sum_{m=1}^n A_{im} (BC)_{mj} \\&= \sum_{m=1}^n A_{im} \sum_{k=1}^p B_{mk} C_{kj} \\&= \sum_{k=1}^p \sum_{m=1}^n A_{im} B_{mk} C_{kj}\end{aligned}$$

$$(AB)C)_{ij} = (A(BC))_{ij} \Leftrightarrow (AB)C = A(BC)$$

$$\begin{aligned}b) (f \circ g) \circ h &= f(g(x)) \circ h \\&= f(g(h(x)))\end{aligned}$$

$$\begin{aligned}f \circ (g \circ h) &= f \circ g(h(x)) \\&= f(g(h(x)))\end{aligned}$$

$$\sin(\arcsin x) = \arcsin(\sin x)$$

8)